

P1.1

1. $I_1 = \int_0^1 \frac{\ln(1+x\sqrt{x})}{x^a} dx$

$\lim_{n \rightarrow \infty} \ln(1+x\sqrt{n}) \sim \ln(x\sqrt{n})$

$a > \frac{3}{2}$ 收敛

$a < \frac{3}{2}$ 发散

2. $I_2 = \int_1^{+\infty} \frac{x^2 + \arctan x}{x^p(x+1)} dx$

$(p + \frac{1}{2}) - 2 > 1$

$p > \frac{5}{2}$ 收敛

P1.2

$I = \int_1^3 \sqrt{x-1} dx$

$x = t^2 + 1$

$x = t^2 + 1$

$dx = 2t dt$

$I = \int_0^{\sqrt{2}} \frac{t}{t^2+1} \cdot 2t dt$

$I = \int_0^{\sqrt{2}} \frac{2t^2}{t^2+1} dt = \int_0^{\sqrt{2}} \frac{2t^2}{t^2+1} dt$

$= \int_0^{\sqrt{2}} 2 dt - \int_0^{\sqrt{2}} \frac{2}{t^2+1} dt$

$= 2t - 2 \arctan(t) \Big|_0^{\sqrt{2}}$

$= 2\sqrt{2} - 2 \arctan(\sqrt{2})$

$$\text{Pr. 1. } L_x = \hat{L}_x + i\hat{L}_y$$

$$\hat{L}_- = \hat{L}_x - i\hat{L}_y$$

$$[\hat{L}_x, \hat{L}_+] = [\hat{L}_x, \hat{L}_x + i\hat{L}_y] = i\hbar\hat{L}_y + i(-i\hbar\hat{L}_x) = \hbar\hat{L}_+$$

$$[\hat{L}_x, \hat{L}_-] = -i\hbar\hat{L}_y + i(-i\hbar\hat{L}_x) = \hbar\hat{L}_-$$

ex)

$$[\hat{L}_x, \hat{L}_-] = [\hat{L}_x + i\hat{L}_y, \hat{L}_x - i\hat{L}_y] = [\hat{L}_x, -i\hat{L}_y] + [i\hat{L}_y, \hat{L}_x] = -i(i\hbar\hat{L}_z) + i(-i\hbar\hat{L}_z) = 2\hbar\hat{L}_z$$

$$\text{ex. 1. 2. } \langle 0, 1, 0 \rangle = \hbar\sqrt{1(1+1)-0(0+1)} \quad |1, 1\rangle = \sqrt{2}\hbar|1, 1\rangle$$

$$L_-|1, 0\rangle = \hbar\sqrt{1(1+1)-0(0+1)} \quad |1, -1\rangle = \sqrt{2}\hbar|1, -1\rangle$$

$$L_x = \frac{i}{2}(\hat{L}_+ + \hat{L}_-) = 0 \quad \langle 1, 1 | 1, 1 \rangle = \frac{\hbar}{2}\langle 1, -1 | 1, -1 \rangle$$

$$\langle \hat{L}_x \rangle = 0$$

$$\langle \hat{L}_x^2 \rangle = \frac{\hbar^2}{2} \langle 1, 1 | 1, 1 \rangle + \langle 1, -1 | 1, -1 \rangle = \hbar^2$$

$$\Delta L_x = \hbar$$

$$\Delta L_y = \hbar$$

$$\hbar^2 \Delta L_z$$

$$\Delta L_x \Delta L_y = \hbar^2$$

$$\langle L_z \rangle = 0$$

$$\Delta L_z$$